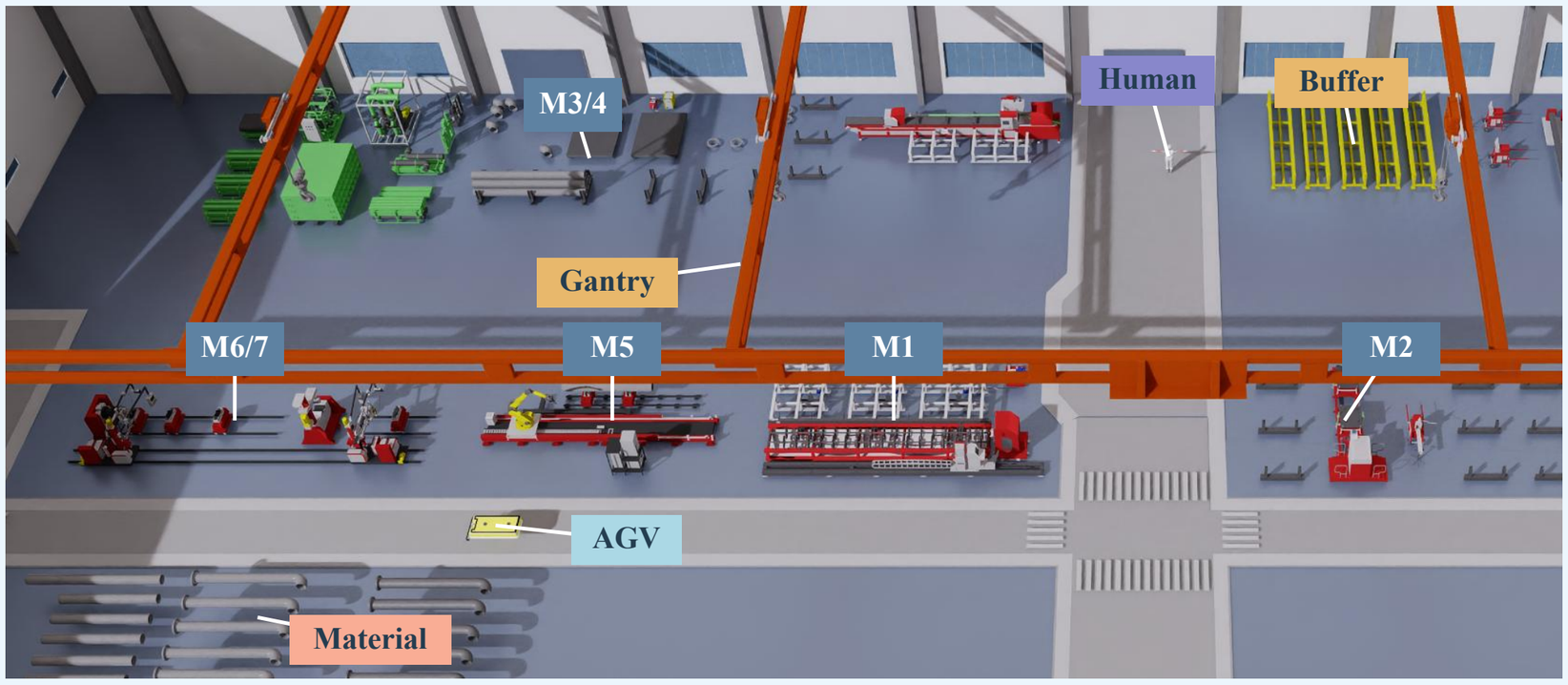
(a) Digital-twin production environment and entities



Machine / workstation identifiers

M1	CNC pipe intersection cutting machine
M2	Large pipe roll-grooving machine
M3	Workbench 0
M4	Workbench 1
M5	Robotic root-welding station
M6	Rotary surface-welding station 0
M7	Rotary surface-welding station 1

M3 / M4 serve both spot welding and painting.
AGV: automated guided vehicle.

(b) Manufacturing and material handling

Manufacturing stages

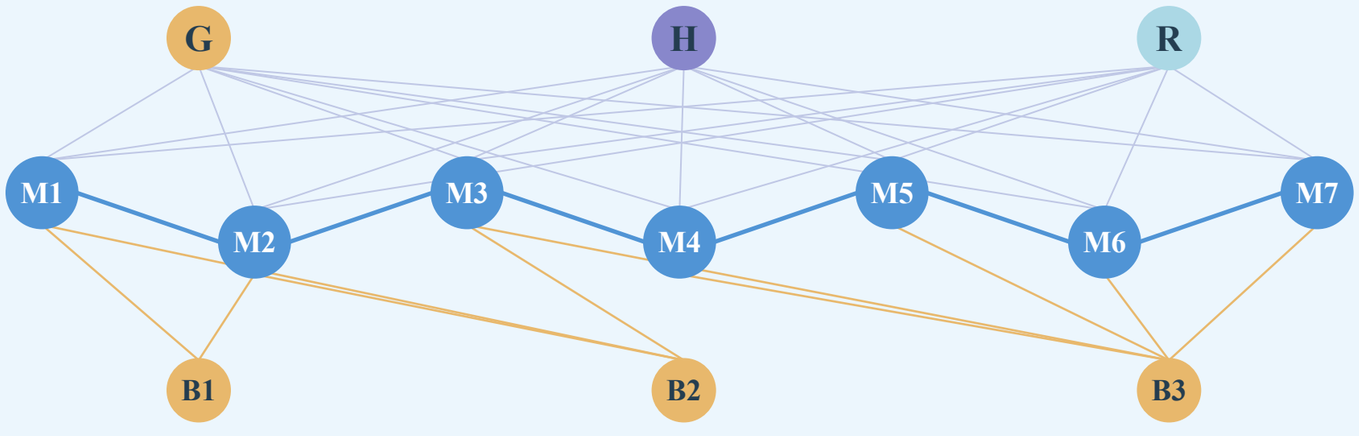
1. **Pipe cutting at M1**
   Deliver raw pipe to M1 → worker operates the cutting machine → cut pipe becomes available for transfer.
2. **Roll grooving at M2**
   Transfer the cut pipe to the large roll-grooving machine → worker operates M2 → grooved pipe proceeds to assembly.
3. **Assembly and spot welding at M3 / M4**
   Supply one pipe, one flange and one elbow → select a workbench → assemble and spot-weld → obtain a semi-finished pipe.
4. **Argon-arc root welding at M5**
   Deliver the semi-finished pipe to the welding robot → worker operates the station → perform root welding.
5. **MIG surface welding at M6 / M7**
   Transfer the root-welded pipe to a selected rotary welding station → perform surface welding → release it for painting.
6. **Rust-proof painting at M3 / M4**
   Return the welded pipe to a workbench → apply rust-proof treatment → produce the finished water pipe.

Shared handling subtasks

7. **Shared processing and outbound handling sequence**
   Worker approaches station → controls machine while it processes → waits for a free gantry → assists loading for outbound transfer.
8. **Material transfer within one gantry zone**
   Worker and gantry approach material → load onto gantry → carry to the destination → place at the target station or storage area.
9. **Material transfer across gantry zones**
   Source gantry loads an AGV → source gantry is released → AGV travels → a free destination gantry unloads and places the material.

(c) Multi-entity structural graph

G: gantry (4) H: human (5) R: AGV (4)



B1: BlackStorage B2: GroundStorage B3: YellowStorage

38 nodes: 7 workstations + 4 gantries + 5 humans + 4 AGVs + 18 buffers.

Graph interpretation

- Blue: workstation backbone links.
- Lavender: human / transport–station links.
- Ochre: buffer–station affinities.

M1–M7 match the workstation list in (a). G, H, R and B1–B3 each represent one node of their resource family.

Edges are undirected structural associations, not task precedence. Repeated nodes, self-loops and within-type transport links are omitted.

Material is visible in the scene but is not a separate graph node. Its effects enter resource features, including supply, waiting and buffer occupancy.